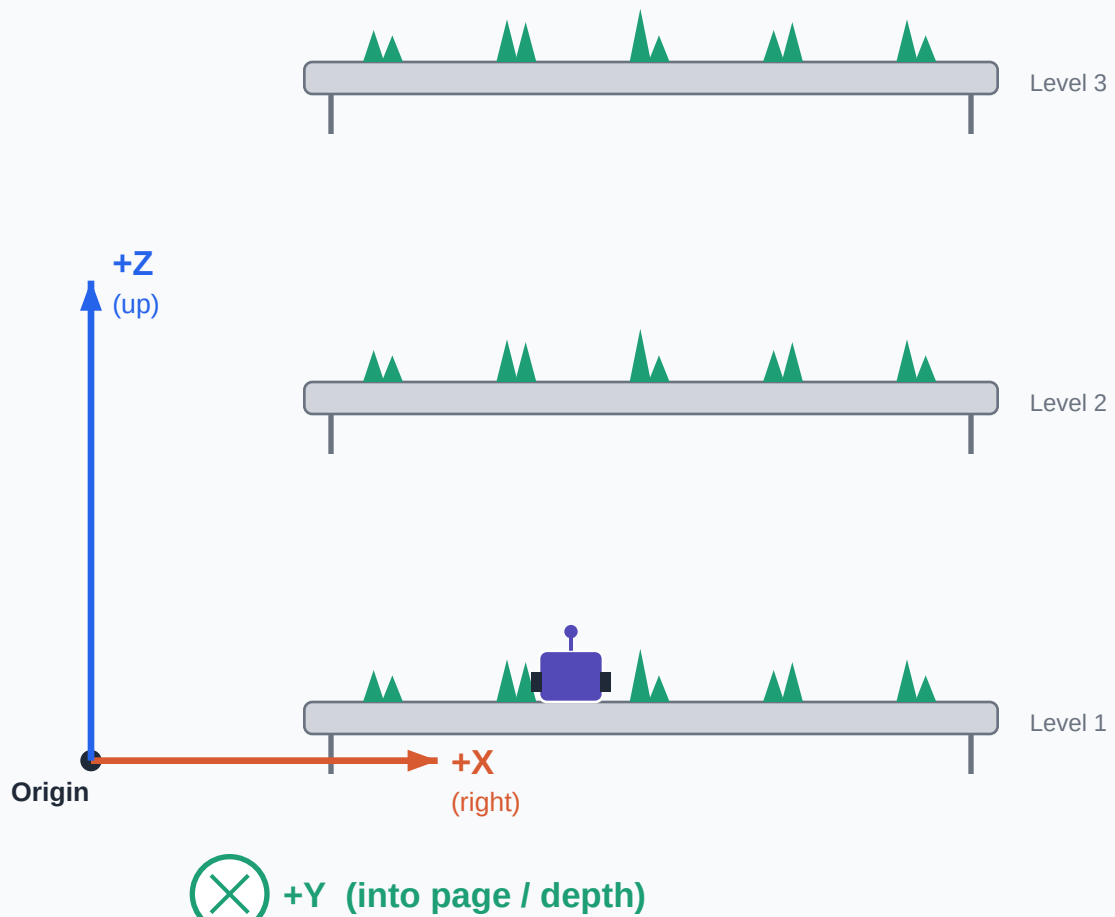


# Coordinate Convention — Vertical Farm Robot

Cross-Section View (standing in front of the farm, looking at it)



## Right-Handed Coordinate System

Curl fingers from +X (right) toward +Y (into page) → thumb points +Z (up) ✓

Standard in robotics, ROS, and engineering. Consistent with heading math.

■ +X left would create a left-handed system — avoid this.

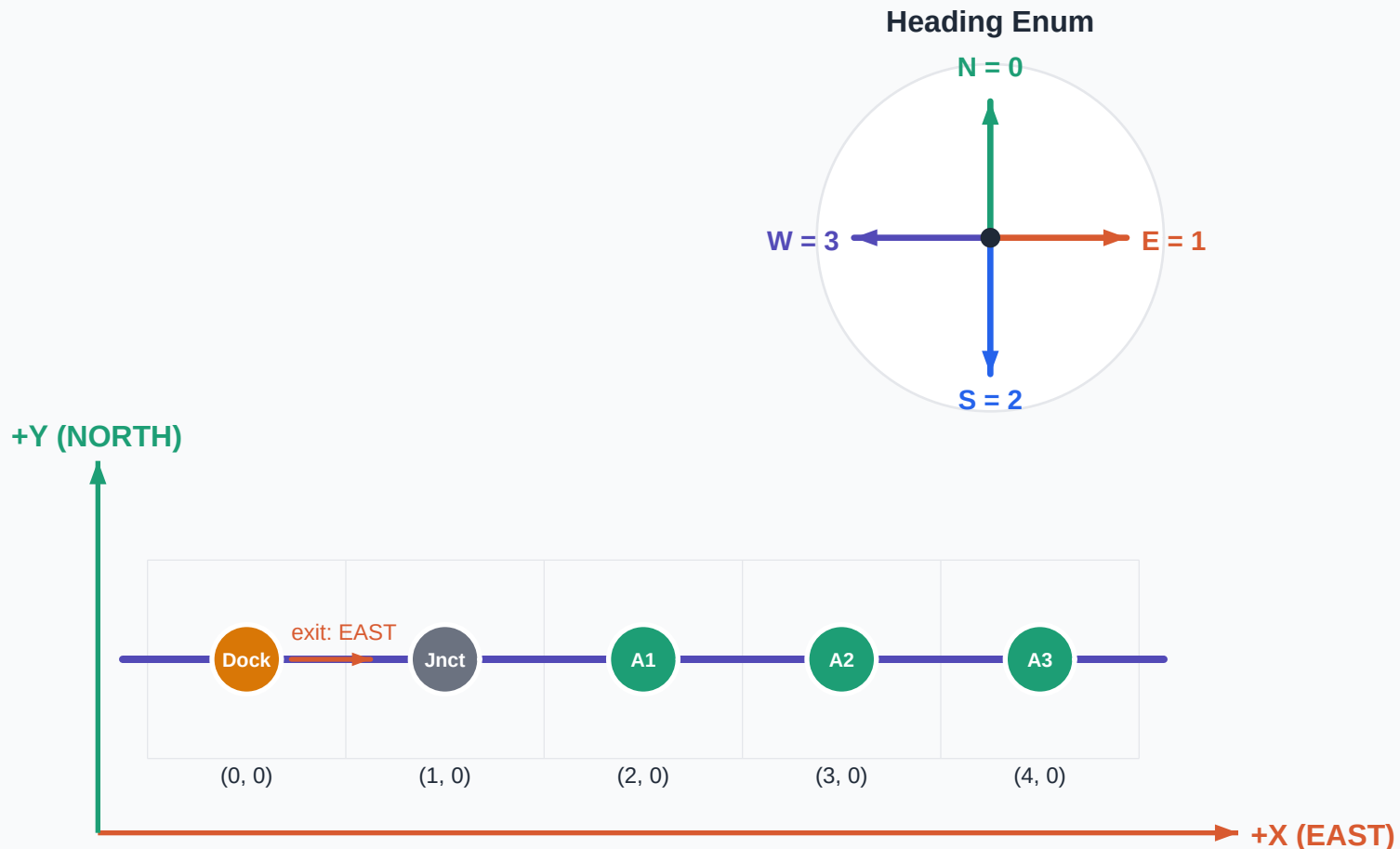
**X** → Along the shelf row (left/right)

**Y** → Depth of the farm (front to back)

**Z** → Vertical height (shelf levels) — handled by elevator, not robot nav

# Floor Navigation — Heading Convention

Top-Down View (looking down at a single level)



## Turn Computation

$$\text{turn} = (\text{desired} - \text{current} + 4) \% 4$$

|   |            |                            |
|---|------------|----------------------------|
| 0 | = STRAIGHT | Continue forward           |
| 1 | = RIGHT    | Turn 90° clockwise         |
| 2 | = U_TURN   | Turn 180°                  |
| 3 | = LEFT     | Turn 90° counter-clockwise |

### Example:

Robot heading EAST (1)  
Next node is NORTH (0)

$$(0 - 1 + 4) \% 4 = 3$$

→ Turn LEFT

## Setting the Starting Heading

Firmware: `robot.heading = EAST; // hardcoded, dock always exits east`

Orchestrator: Uses `exit_heading` from `topology.json` dock node

After first move, heading is maintained automatically by `computeHeading(from, to)`